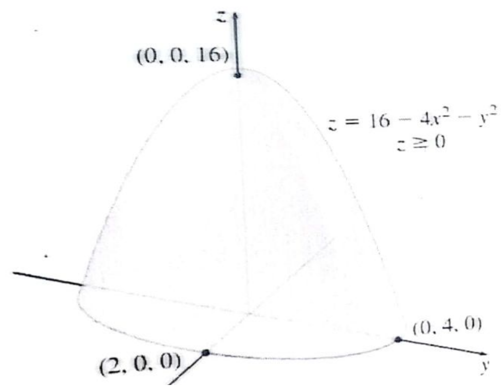
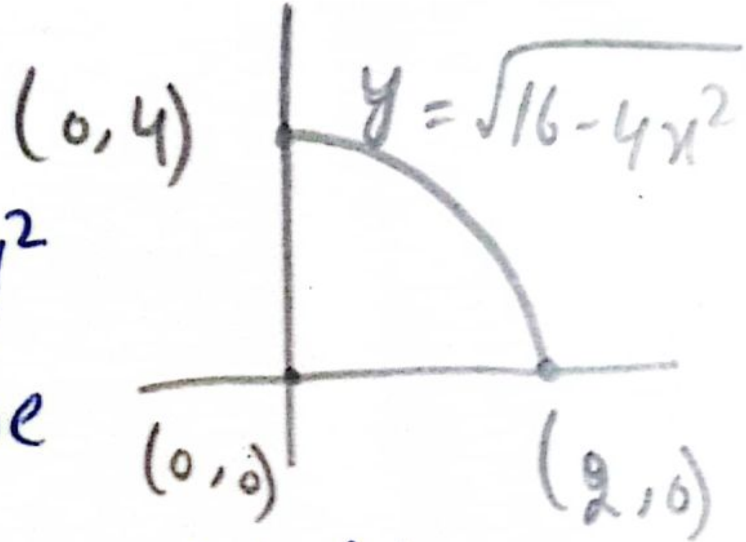
Q1: Setup an iterated integral (triple integral) giving the volume V of the solid E in the first octant that is enclosed by the paraboloid $z = 16 - 4x^2 - y^2$ and the $xy - plane$. Sketch the projection of E onto $xy - plane$. Also, give the integrands and bounds, but do not evaluate. **[5 marks]**



Sol upper surface is the paraboloid $z = z_2(x,y) = 16 - 4x^2 - y^2$ & lower surface is the plane $z = z_1(x,y) = 0$. Region R in xy-plane



is enclosed by x-axis, y-axis & part of ellipse $4x^2+y^2=16$, $x \geq 0, y \geq 0$, & z_1 & z_2 are cont on R. So E is xy-simple. The region R is both x-simple & y-simple. We choose to use the iterated integral for an x-simple region, so $0 \leq y \leq \sqrt{16-4x^2}$ & $0 \leq x \leq 2$.

$$V = \int_a^b \int_{y_1(x)}^{y_2(x)} \int_{z_1(x,y)}^{z_2(x,y)} dz\,dy\,dx \Rightarrow \int_0^2 \int_0^{\sqrt{16-4x^2}} \int_0^{16-4x^2-y^2} dz\,dy\,dx$$

Q2: An agricultural sprinkler distributes water in a circular pattern of radius 100 feet. It supplies water to a depth of e^{-r} feet per hour at a distance of r feet from the sprinkler.

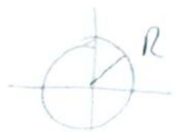
a) If $0 < R \leq 100$, what is the total amount of water supplied per hour to the region inside the circle of radius R centered at the sprinkler?
b) Determine an expression for the average amount of water per hour per square foot supplied to the region inside the circle of radius R. **[10 marks]**

R centered at sprinkler?

$$\int_0^{2\pi}\int_0^{R} e^{-r} r\,dr\,d\theta$$

$$\left(\int_0^{2\pi} d\theta\right)\left(\int_0^{R} e^{-r} r\,dr\right)$$

$$2\pi \int_0^{R} e^{-r} r\,dr$$



$0 \le r \le R$

$0 \le \theta \le 2\pi$

Let $u = r$, $v = e^{-r}$

$$2\pi\left[r\int e^{-r}dr - \left[\int\left(\frac{d}{dr}\cdot r\right)\left(\int e^{-r}dr\right)\right]dr\right]_0^R$$

$$2\pi\left[r(-e^{-r}) - \left[\int 1\cdot(-e^{-r})\right]dr\right]_0^R$$

$$2\pi\left[r(-e^{-r}) + (-e^{-r})\right]_0^R$$

$$2\pi\left[-e^{-R}(1+R)+1\right]$$

$$2\pi\left[1 - \frac{R+1}{e^R}\right]$$

$$2\pi - 2\pi R e^{-R} - 2\pi e^{-R}$$

If $R = 100$

$$2\pi\left[1 - \frac{100+1}{e^{100}}\right] \Rightarrow -202\pi e^{-100} + 2\pi$$

(b) Determine an expression for the average amount of water per hr per sq ft supplied to the region inside circle of radius R.

Sol

$$\frac{1}{\pi R^2}\cdot 2\pi\left[1 - \frac{R+1}{e^R}\right]$$

$$\frac{2}{R^2}\left[1 - \frac{R+1}{e^R}\right]$$

Q3: In neural networks and machine learning, deriving efficiently is vital. Traditional methods are computationally heavy, but automatic differentiation offers a swift alternative. As a computer science student, your task is to apply automatic differentiation to compute the partial derivatives of $z = \sin(x_1 + x_2) + x_2^2 x_3$. **[15 marks]**

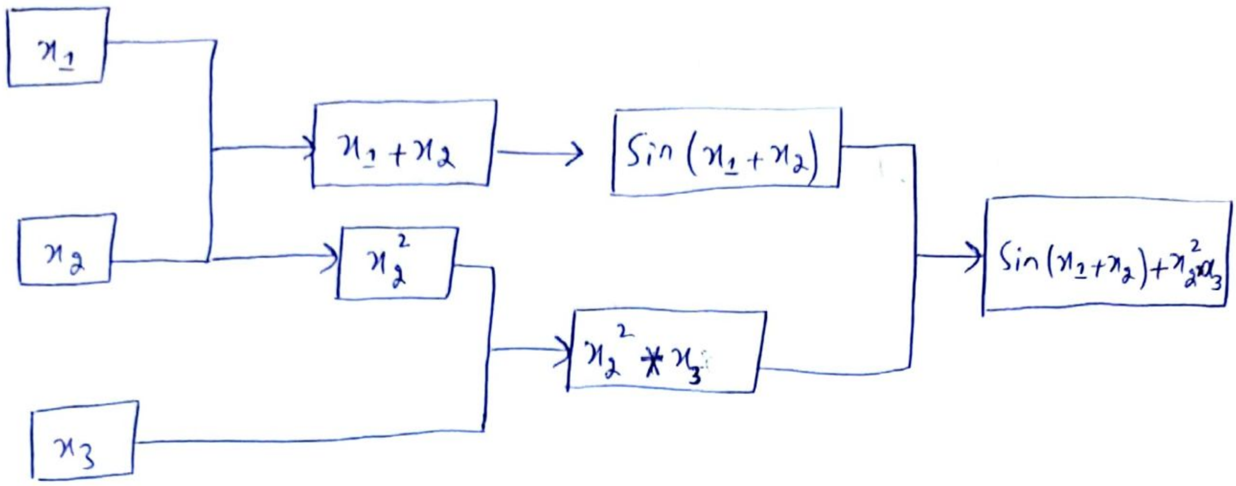
i- Create the computational graph.

ii- Find $\frac{\partial z}{\partial x_1}, \frac{\partial z}{\partial x_2}$, and $\frac{\partial z}{\partial x_3}$ using reverse mode automatic differentiation.

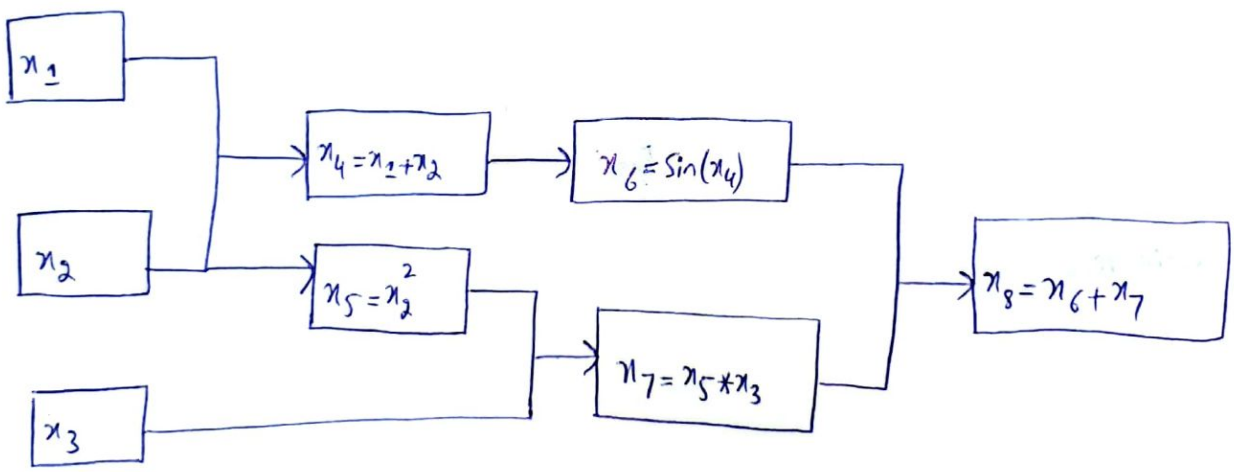
iii- Directly calculate $\frac{\partial z}{\partial x_1}, \frac{\partial z}{\partial x_2}$, and $\frac{\partial z}{\partial x_3}$ and compare your answers.

Q:- $z = \sin(x_1 + x_2) + x_2^2 x_3$

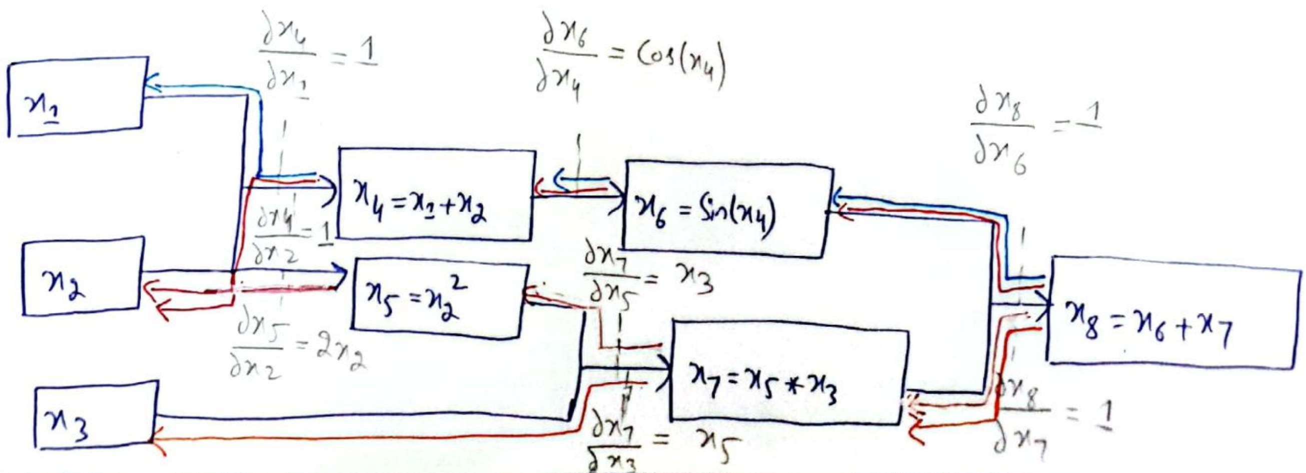
Sol Computational Graph



Forward Pass



Backward Pass



$$* \frac{\partial z}{\partial x_1} = \frac{\partial x_8}{\partial x_6} \times \frac{\partial x_6}{\partial x_4} \times \frac{\partial x_4}{\partial x_1}$$

$$\frac{\partial z}{\partial x_1} = 1 \times \cos(x_4) \times 1$$

$$\frac{\partial z}{\partial x_1} = \cos(x_4) \Rightarrow \frac{\partial z}{\partial x_1} = \cos(x_1 + x_2)$$

$$* \text{ Path } 2 = \frac{\partial x_8}{\partial x_7} \times \frac{\partial x_7}{\partial x_5} \times \frac{\partial x_5}{\partial x_2}$$

$$\text{Path } 2 = 1 \times x_3 \times 2x_2$$

$$\text{Path } 2 = 2x_2 x_3$$

$$\frac{\partial z}{\partial x_2} = \text{Path } 1 + \text{Path } 2$$

$$\text{Path } 1 = \frac{\partial x_8}{\partial x_6} \times \frac{\partial x_6}{\partial x_4} \times \frac{\partial x_4}{\partial x_2}$$

$$\text{Path } 1 = 1 \times \cos(x_4) \times 1$$

$$\text{Path } 1 = \cos(x_4)$$

$$\frac{\partial z}{\partial x_2} = \cos(x_4) + 2x_2 x_3 \Rightarrow \frac{\partial z}{\partial x_2} = \cos(x_1 + x_2) + 2x_2 x_3$$

$$* \frac{\partial z}{\partial x_3} = \frac{\partial x_8}{\partial x_7} \times \frac{\partial x_7}{\partial x_3}$$

$$\frac{\partial z}{\partial x_3} = 1 \times x_5$$

$$\frac{\partial z}{\partial x_3} = x_5 \Rightarrow \frac{\partial z}{\partial x_3} = x_2^2$$

Direct derivative

$$\frac{\partial z}{\partial x_1} = \cos(x_1 + x_2)$$

$$\frac{\partial z}{\partial x_2} = \cos(x_1 + x_2) + 2x_2 x_3$$

$$\frac{\partial z}{\partial x_3} = x_2^2$$

Start writing here

Q4 $f(x,y) = x^2 + y^2 - 4x - 6y + 13$

Sol $f_x(x,y) = 2x - 4 \Rightarrow f_{xx}(x,y) = 2 \quad f_{xy}(x,y) = 0$

$f_y(x,y) = 2y - 6 \Rightarrow f_{yx}(x,y) = 0 \quad f_{yy}(x,y) = 2$

Then $\nabla f(x,y) = \begin{bmatrix} 2x-4 \\ 2y-6 \end{bmatrix} \qquad H = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

Let X_0 be the initial approximation of the form $(a,0)$

Let $X_0 = (3,0)$

then

$S_0 = \nabla f(X_0) = \begin{bmatrix} 2(3)-4 \\ 2(0)-6 \end{bmatrix} = \begin{bmatrix} +2 \\ -6 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\lambda_0 = \frac{S_0^T S_0}{S_0^T H S_0} = \frac{\begin{bmatrix} 2 & -6 \end{bmatrix} \begin{bmatrix} 2 \\ -6 \end{bmatrix}}{\begin{bmatrix} 2 & -6 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ -6 \end{bmatrix}}$$

$$= \frac{[4+36]}{\begin{bmatrix} 2 & -6 \end{bmatrix} \begin{bmatrix} 4 \\ -12 \end{bmatrix}} = \frac{[40]}{[8+72]} = \frac{[40]}{[80]}$$

$$\boxed{\lambda_0 = \frac{1}{2}}$$

then $X_1 = X_0 - \lambda_0 S_0$

$$= \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \frac{1}{2}\begin{bmatrix} 2 \\ -6 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

<u>Step II:</u>

$$S_1 = \nabla f(X_1) = \begin{bmatrix} 2(2)-4 \\ 2(3)-6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus Gradient vector becomes zero.

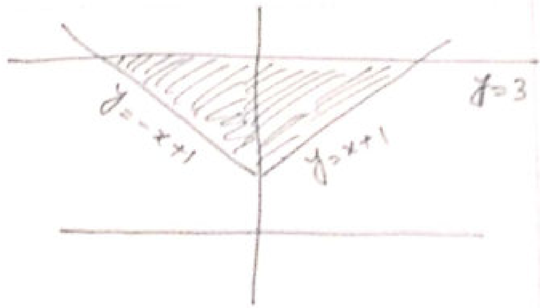
Hence at $X_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$ or $(2, 3)$ there is a minima

of $f(x, y) = x^2 + y^2 - 4x - 6y + 13$.

Ans

Question 5 $\iint_R (2x - y^2)\, dA$ $y = -x+1$, $y = x+1$, $y = 3$

Sol It is better to integrate first by 'x'



$$I = \int_1^3 \int_{1-y}^{y-1} (2x - y^2)\, dx\, dy$$

$$= \int_1^3 \left[x^2 - y^2 x\right]_{1-y}^{y-1} dy$$

$$= \int_1^3 (2y^2 - 2y^3)\, dy$$

$$= \left[\frac{2y^3}{3} - \frac{y^4}{2}\right]_1^3 = \frac{-68}{3} \text{ Ans}$$

Q6

$x^2+y^2-z=0$ and $3x^2+2y^2+z^2-9=0$

$F(x,y,z) = x^2+y^2-z$ $G(x,y,z) = 3x^2+2y^2+z^2-9$

$\nabla F(x,y,z) = 2xi+2yj-k$

$\nabla G(x,y,z) = 6xi+4yj+2zk$

$\nabla F(1,1,2) = 2i+2j-k$

$\nabla G(1,1,2) = 6i+4j+4k$

$$\nabla F \times \nabla G = \begin{vmatrix} i & j & k \\ 2 & 2 & -1 \\ 6 & 4 & 4 \end{vmatrix}$$

$= 12i - 14j - 4k$

$= 6i - 7j - 2k$

so

$x = 1+6t$, $y = 1-7t$, $z = 2-2t$